4.3.5 Wrap-Up: Reflection

1. Check the box that describes your understanding of each learning target.

- ☐ I can identify transformations that preserve distance and I am ready for the next lesson.
- ☐ I need some help identifying transformations that preserve distance. One area in which I need help is

  ____.

- ☐ I need more support in knowing how to identify transformations that preserve distance. I am concerned I will not be ready to move on. One area in which I need help is

  ____.

- ☐ I can specify a sequence of transformation that will map a given figure onto another congruent figure.
- ☐ I need some help specifying a sequence of transformation that will map a given figure onto another congruent figure. One area in which I need help is

  ____.

- ☐ I need more support in knowing how to identify transformations that preserve distance. I am concerned I will not be ready to move on. One area in which I need help is

  ____.

Unit 4, Lesson 4: Written Descriptions of a Sequence of Transformations

Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

Learning Targets

- ☐ I can draw an image given a pre-image and a written description of a sequence of transformations on a coordinate plane.
- ☐ I can describe a transformation on the coordinate plane using written descriptions.

4.4.1 Warm-Up: Notice and Wonder

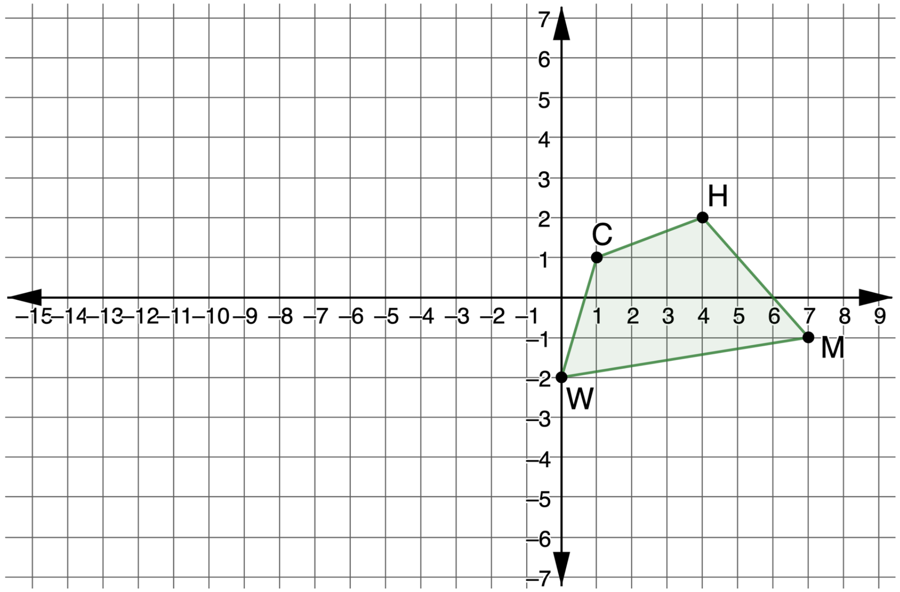
A kaleidoscope is an instrument that contains loose bits of colored material, usually glass, between two flat plates and two plane mirrors. The plates can be moved so the colored material changes position. The mirrors then reflect the material in an endless variety of patterns.

1. What do you notice about the pattern from the kaleidoscope in the picture?

2. What do you wonder about the pattern?

4.4.2 Collaboration: Drawing the Image

1. Quadrilateral $CHMW$ is shown on the coordinate grid.



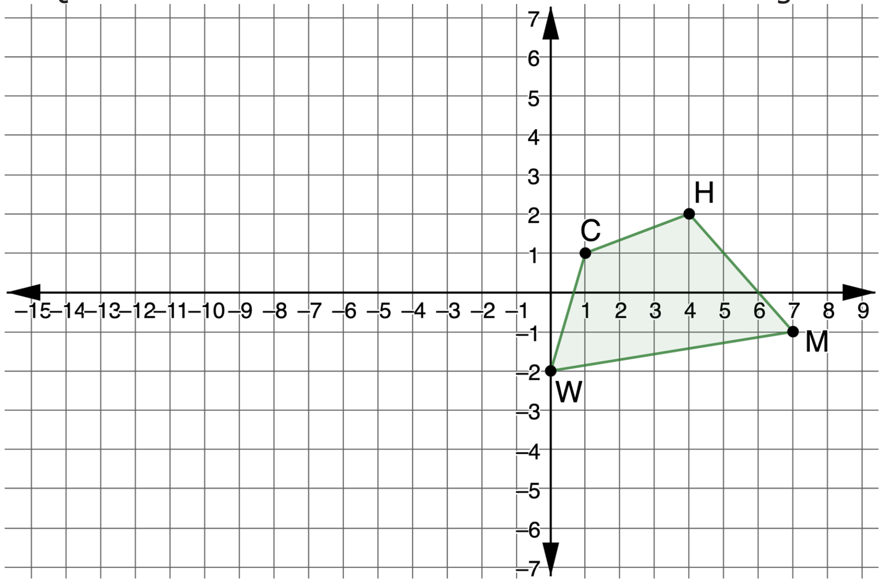
   a. Draw the quadrilateral that is the result of a reflection across the y-axis and then translated left 3 and up 1. Label the new quadrilateral $BVXN$.

   b. Check your graph with your partner. Make any necessary adjustments.

A ***sequence of transformations***, or a composition of transformations, is when two or more transformations are combined.

c. Work with your partner to explain why quadrilateral $BVXN$ is a result of a sequence of transformations.

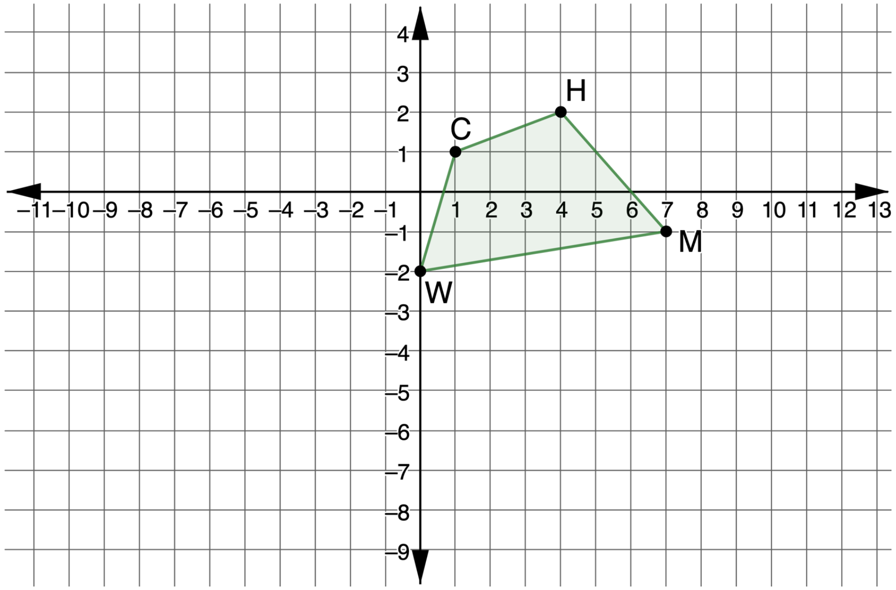
2. Quadrilateral $CHMW$ is shown on the coordinate grid.



a. Draw the quadrilateral that is the result of a rotation of $180°$ counterclockwise about the origin and then reflected across the x-axis. Label the new quadrilateral $KDPL$.

b. Check your graph with your partner. Make any necessary adjustments.

3. Quadrilateral $CHMW$ is shown on the coordinate grid.



a. Draw the quadrilateral that is the result of a rotation of $90°$ clockwise about the point W and then translated left 2 and up 5. Label the new quadrilateral $RYFG$.

b. Check your graph with your partner. Make any necessary adjustments.

4. For each sequence of transformations, summarize your mistakes and what you learned from them.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

a. A reflection across the y-axis and then translated left 3 and up 1

b. A rotation of $180°$ counterclockwise about the origin and then reflected across the x-axis

c. A rotation of $90°$ clockwise about the point W and then translated left 2 and up 5

4.4.3 Guided Instruction: Using Coordinates

The Effect of Transformations on Coordinates

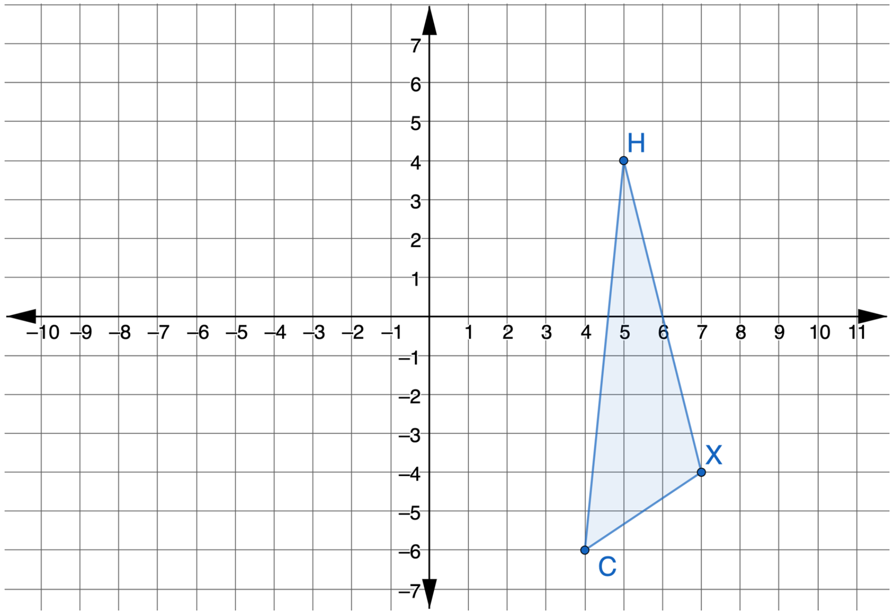
Rotations Clockwise About the Origin		
Rotations of $90°$	**Rotations of $180°$**	**Rotations of $270°$**
Change the x-coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the y-coordinate to its opposite, then switch the coordinates

Rotations Counterclockwise About the Origin		
Rotations of $90°$	**Rotations of $180°$**	**Rotations of $270°$**
Change the y-coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the x-coordinate to its opposite, then switch the coordinates

Translations	
Right	**Left**
Add to the x-coordinate	Subtract from the x-coordinate
Up	**Down**
Add to the y-coordinate	Subtract from the y-coordinate

Reflections	
x-axis	**y-axis**
The y-coordinate changes to its opposite.	The x-coordinate changes to its ______________.
$y = x$	**$y = -x$**
The coordinates switch.	The coordinates switch and become their opposite.

1. Triangle CHX is shown on the coordinate grid.



Triangle CHX

Vertex	x	y
C	4	−6
H	5	4
X	7	−4

a. Reflect ΔCHX across the y -axis then rotate $90°$ clockwise about the origin. Show both $\Delta C'H'X'$ and $\Delta C''H''X''$.

b. Complete the table of values for each vertex.

Triangle $C'H'X'$

Vertex	x	y
C'		
H'		
X'		

Triangle $C''H''X''$

Vertex	x	y
C''		
H''		
X''		

c. Discuss with your partner how the tables show the effect of transformations on coordinates.

One strategy to apply transformations is to use tables of ordered pairs and then to plot the ordered pairs.

2. The table of values shows the coordinates of the vertices of quadrilateral $WBKT$.

Quadrilateral $WBKT$

Vertex	x	y
W	−17	−5
B	−11	−5
K	−4	−11
T	−14	−16

a. Complete the table of values for the given sequence of transformations of quadrilateral $WBKT$.

Rotate quadrilateral $WBKT$ $180°$ counterclockwise about the origin, then translate it right 7 down 4.

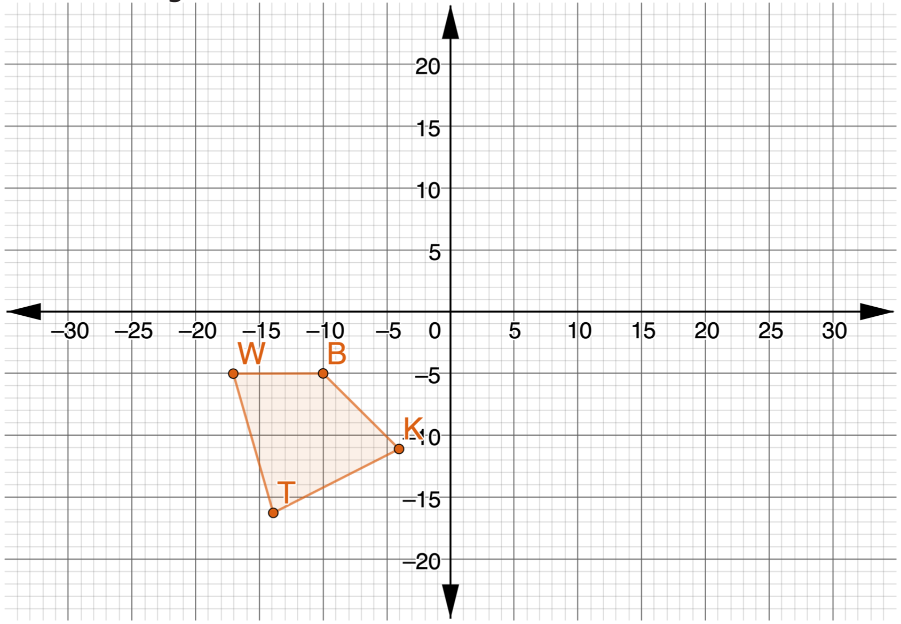
Quadrilateral $W'B'K'T'$

Vertex	x	y
W'		
B'		
K'		
T'		

Quadrilateral $W''B''K''T''$

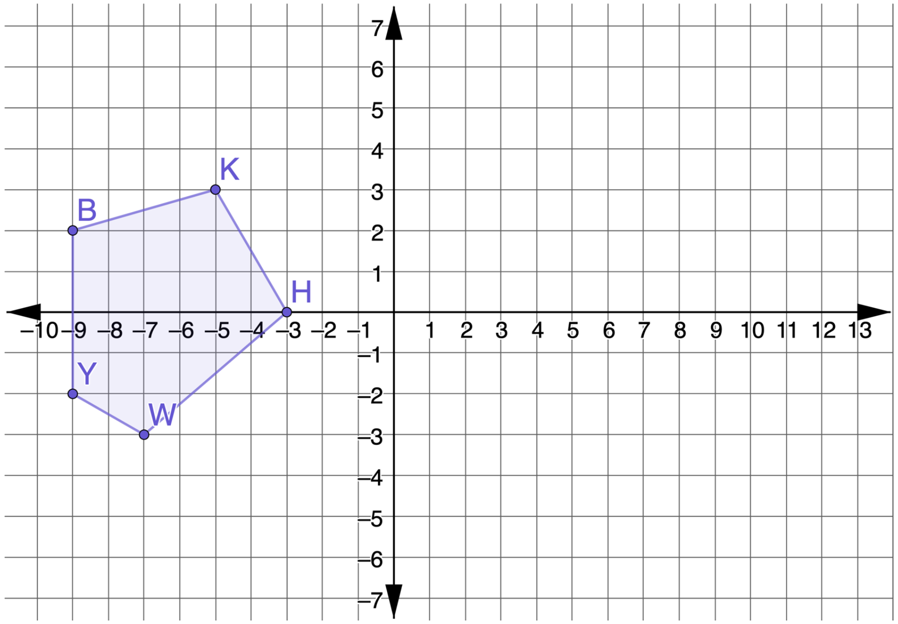
Vertex	x	y
W''		
B''		
K''		
T''		

b. Quadrilateral $WBKT$ is shown on the coordinate grid. Show the sequence of transformations on the coordinate grid by rotating quadrilateral $WBKT$ $180°$ counterclockwise about the origin, then translating it right 7 and down 4.



4.4.4 Your Turn!

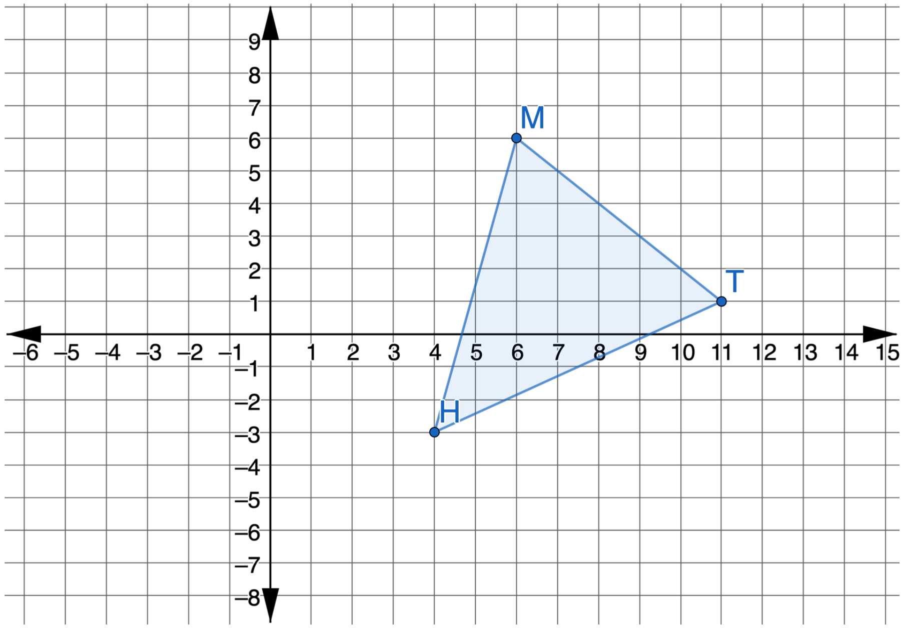
1. Polygon $BKHWY$ is shown on the coordinate grid.



On the coordinate grid, translate polygon $BKHWY$ right 5 and down 3. Then, reflect the result across the line $x = 4$. Label the resulting polygon $B''K''H''W''Y''$.

4.4.5 Wrap-Up: Ticket Out the Door

1. Triangle HMT is shown on the coordinate grid.



Draw ΔBWF, which is the result of translating ΔHMT left 3 and up 3, then rotating the result $90°$ clockwise about the origin.

Unit 4, Lesson 5: Algebraic Descriptions of a Sequence of Transformations

Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

Learning Targets

- ☐ I can draw an image given a pre-image and an algebraic description of a sequence of transformations on a coordinate plane.
- ☐ I can write a transformation on the coordinate plane using coordinate notation with algebraic descriptors.